

Weekly Report: Week 6

2026-09-02

This report covers only what is new since Week 5. The pipeline, the elicitation and selection-rule comparisons, the Fodors–Zagat and DBLP–ACM result tables, and the three-draw DBLP–ACM repeat are as reported there and are not restated.

Nine runs were executed for this report, totalling 344,888 matcher calls with **zero API errors**. All use the matcher `deepseek-v4-flash-0731` and the embedding model `bge-large-en-v1.5`.

1 Hyperparameters

Requested for this report, and previously only recoverable from the run summaries. Everything below is held fixed across every run in Weeks 5 and 6 unless a result explicitly varies it.

Stage	Parameter	Value
Embedding	model encoding	BAAI/bge-large-en-v1.5 per column, concatenated, L2-normalised
Blocking	scheme tables / planes / flips top- k per Table-A record	multi-probe LSH, random hyperplanes 15 / 8 / 1 (DBLP-ACM), 10 / 4 / 1 (Amazon-Walmart) 5 (DBLP-ACM), 4 (Amazon-Walmart)
Profiling	sampler sample size elicitation concurrency	diversity 50 candidate pairs decisive (ordinal where stated) 4 workers
Transfer	predictor retriever fusion	Ridge regression over 6 features per attribute cosine k -NN, $k = 5$ weighted_average , 0.5 / 0.5
Selection	rule k bounds cumulative target ratio λ required roles blank-field handling attribute order in prompt	gap (no threshold) min = 1, max = 5 0.8 (only for the cumulative rule) 1.5 (only for the ratio rule) identity drop (Section 4) canonical (table-column order)
Matcher	model temperature max tokens reasoning concurrency	deepseek/deepseek-v4-flash-0731 0 256 disabled 8 workers
Seeds	pipeline / blocking candidate slice (repeats)	42 / 42 20,000 pairs, seed 7

Table 1: Hyperparameters. Attribute *roles* are assigned per dataset from the column name: **title** and **modelno** are **identity**, **authors** and **brand** **strong_support**, **venue**, **year** and **category** **context**, and the remaining Amazon-Walmart columns **other**. Only **identity** is required, which is what forces at least one identifying field into every selection.

2 The Matcher Drifts, and the Drift Is Now Measured Directly

Week 5 inferred provider drift from the fact that F1 levels moved while the pipeline did not. It can be measured without that inference. The August and late-August response caches share **69,087 Amazon-Walmart prompts that are identical byte for byte**. Replaying the same prompt to the same model slug on the two dates:

Identical prompts compared	69,087
Answered Yes in August	1,589
Answered Yes in late August	3,250
Moved No $\rightarrow$ Yes	2,269
Moved Yes $\rightarrow$ No	608
Disagreement rate	4.16%

Table 2: The same input, the same model slug, two dates. The imbalance is 3.7 to 1 toward Yes, so this is a directional shift and not sampling noise.

The consequence is visible in one configuration held constant across three dates (Step 1.5, decisive, `gap`, $k \geq 1$, all 88,296 Amazon–Walmart pairs):

Date	Precision	Recall	F1
19 August	0.4762	0.7962	0.5959
20 August	0.4986	0.7997	0.6142
31 August	0.2215	0.8093	0.3478

Table 3: Identical code, identical prompts, identical model slug. Precision more than halves; recall does not move, because a matcher answering Yes more often loses precision and keeps recall.

The routing behind this is now observed rather than inferred. Until this week nothing in the recorded response could name the cause, so provider variation was a hypothesis. OpenRouter returns a non-standard `provider` field on each completion, and the pipeline now records it per call, into the response cache and into `final_results.csv`, with a per-run histogram in the run summary. Forty calls issued back to back against the single slug `deepseek/deepseek-v4-flash-0731` were served by **eleven distinct providers**: OpenInference (22%), DigitalOcean (18%), Ambient and Relace (12% each), DeepInfra (10%), Inceptron (8%), Together and StreamLake (5% each), and Reka, Fireworks and Sail Research (one call each). One slug is therefore not one model in any operational sense, and no run of this pipeline before this week can say which mixture produced it.

No comparison may place its two arms on different dates. Every run reported below was executed inside a single window for this reason.

Amazon–Walmart is the noisier benchmark, not the calmer one. Its seven times larger pair count might be expected to average random flips away. Two runs of no selection over an identical fixed slice of 20,000 pairs, both on 27 August, disagree on 1,040 of them — **5.20%**, above DBLP–ACM’s 4.82% — and on **0.078 F1**, 0.3299 against 0.4083. Several differences reported for this dataset in Week 5 are smaller than that.

3 Repeating the Amazon–Walmart Comparison Dissolves It

The Week 5 table separates no selection from adaptive selection by 0.023 F1, against a run-to-run spread of 0.078. The comparison was repeated three times per arm on a fixed random slice of 20,000 candidate pairs (seed 7, containing 233 of the 1,143 ground-truth matches), all six runs inside one day, each writing its own response cache so that every draw is independent rather than a replay.

Configuration	Draw 1	Draw 2	Draw 3	Mean	Range
No selection (full)	0.3442	0.4311	0.3953	0.3902	[0.3442, 0.4311]
Step 1.5, decisive (gap)	0.3204	0.3820	0.3945	0.3656	[0.3204, 0.3945]

Table 4: Amazon–Walmart F1 over three independent draws per arm, 31 August–1 September. Levels are not comparable with the Week 5 table, which scores over all 88,296 pairs and was run three weeks earlier.

The ranges overlap almost completely, and the paired tests do not agree on a sign. Pairing the draws in order gives no selection uniquely correct on 755 pairs against 725 ($p = 0.45$), then 588 against 514 ($p = 0.028$), then 538 against 594 ($p = 0.10$) — a tie, a win for no selection, and a lead for adaptive selection, from three repetitions of one experiment. The two configurations are not distinguishable on this dataset, and the 0.023 F1 advantage reported in Week 5 is a property of which draw was taken. **That ordering is withdrawn.**

Two quantities do survive repetition.

	No selection	Step 1.5, decisive
Recall (range over 3 draws)	[0.9313, 0.9528]	[0.8670, 0.8755]
Precision (range over 3 draws)	[0.2111, 0.2785]	[0.1965, 0.2554]
Pipeline tokens (mean)	20,607,108	5,186,455

Table 5: Recall separates cleanly and in the same direction every draw — a consistent cost of roughly 0.074 — while precision overlaps, so nothing is bought back. Cost separates by far more: a 74.8% reduction, of which profiling consumes 55,448 tokens.

The defensible claim on this schema is therefore narrower than Week 5 states it. Pair-adaptive selection removes three quarters of the token cost with no measurable change in F1 and a reliable loss of recall. It is a cost–quality trade, not an accuracy result, and it stands in deliberate contrast to DBLP–ACM, where the same configuration separates from no selection by 0.186 with non-overlapping ranges.

4 Blank Fields: Removing Them and Hiding Them Are Two Different Things

The evidence mask of Week 5 zeroes an attribute’s importance when either record leaves it blank, which bars it from being *selected*. As implemented it also keeps the field out of the prompt entirely. Those are separable operations, and the distinction was raised by the supervisor: a matcher shown **modelno** as an empty field can read it as unknown, whereas one never shown the field cannot distinguish a missing value from a schema without that column.

A second mode, **keep**, bars the attribute from selection exactly as before but appends it, blank, to the prompt. Selection is byte-identical under both, so the runs differ only in what the matcher reads. Both were run back to back on one day over all 88,296 pairs, sharing a response cache so that the 62,770 pairs whose prompt is unchanged replay rather than being re-drawn.

Blank fields in the prompt	Precision	Recall	F1	TP	FP
Removed (drop)	0.2215	0.8093	0.3478	925	3,251
Shown (keep)	0.2017	0.8119	0.3231	928	3,674
Shown only when blank on both sides	0.2251	0.8075	0.3521	923	3,177

Table 6: Amazon–Walmart, all three runs 31 August–1 September. Levels sit far below the same configuration’s 0.6142 in Week 5 for the reason given in Section 2; only the within-table comparison is meaningful.

Showing every blank field costs 423 false positives to buy 3 true ones, and loses the paired test 1,155 uniquely correct pairs to 737 ($p = 6.2 \times 10^{-22}$) over the 25,526 pairs whose prompt differs. **Splitting those pairs by which kind of blank was restored shows that the single answer was hiding two opposite ones.**

Restored attribute was	Pairs	Gold matches	drop right	keep right	p
blank on one side	12,894	242 (1.88%)	999	504	7.5×10^{-38}
blank on both	12,560	14 (0.11%)	157	233	1.1×10^{-4}

Table 7: The two kinds of missingness carry opposite signal.

The gold labels explain the reversal without reference to the matcher. Against a base match rate of 1.18% over the candidate pairs, a field blank on exactly one side is mildly *pro*-match, while a field blank on both is strongly anti-match:

Attribute	Both sides present	One side blank	Both sides blank
<code>modelno</code>	1.26%	1.88%	0.11%
<code>shortdescr</code>	1.10%	1.97%	0.35%
<code>dimensions</code>	1.27%	2.05%	0.52%
<code>longdescr</code>	1.17%	2.74%	0.00%
<code>brand</code>	1.18%	1.85%	0.00%

Table 8: Match rate among the 88,296 Amazon–Walmart candidate pairs, base rate 1.18%. A sparsely catalogued record is more often a genuine second listing of the same product than a different product described in full, which is why a one-sided blank is not evidence against a match. A field neither record carries is: `modelno` at 0.11% is nearly ten times below base. No LLM calls were needed to produce this table.

Showing a half-filled field invites agreement it has not earned; showing that neither record has the field is a genuine reason to doubt. A third mode keeping only the second case wins the paired test against `drop`, 233 to 157 over 12,632 affected pairs ($p = 1.4 \times 10^{-4}$), and is implemented as `--evidence-mask-mode both.blank`.

This is reported as mechanism, not as an improvement, and drop remains the default everywhere else. The margin is 0.0043 F1 against the 0.078 spread two identical runs exhibit on this dataset; it was found by inspecting the evaluation set’s labels with no held-out split; and on DBLP–ACM the mask binds on 5 of 13,080 pairs, so the distinction cannot matter there. What the split contributes is the finding that missingness is not one signal but two, pointing in opposite directions — which the current feature vector represents identically.

5 What Changes in the Thesis Narrative

The Amazon–Walmart claim is reduced from accuracy to cost. The dataset now supports one statement: pair-adaptive selection removes 74.8% of the token cost with no measurable change in F1 and a repeatable 0.074 loss of recall. The 0.023 advantage claimed for no selection, and the implication that adaptive selection trails there, are both withdrawn.

Having two datasets reach different conclusions is the stronger result. DBLP–ACM separates by 0.186 with disjoint ranges; Amazon–Walmart does not separate at all. Together they bound where the method applies — aggressive condensation wins outright where one or two attributes can carry the decision, and degrades to a cost saving where the evidence is spread across a wide, sparsely populated schema. A report claiming a win on both would be harder to defend, not easier.

Non-reproducibility of a commercial LLM matcher is a finding in its own right, and its mechanism is now identified. Two identical runs disagree on 5.2% of pairs and 0.078 F1; 69,087 identical prompts replayed a fortnight later move from 1,589 Yes answers to 3,250; and forty consecutive calls to one model slug are served by eleven different providers. Any result this pipeline obtains through a hosted model API inherits this, and the provider is not recoverable after the fact from a run that did not record it — as none of this project’s runs before this week did. Whether related work reports this is not something this report has checked, and no claim is made about it. Condensation halves the variance (Week 5), which is a second argument for the method that has nothing to do with accuracy.

6 Limitations Added or Changed This Week

One headline comparison did not survive repetition. Both headline rows are now repeated three times on each dataset. On DBLP–ACM the ordering is stable and the ranges disjoint; on Amazon–Walmart the ranges overlap and the sign of the paired test changes between draws. Every remaining row — the whole of Fodors–Zagat, the six other Amazon–Walmart configurations, and the rule sweep — is still a single draw, and the Amazon–Walmart spread of 0.078 F1 exceeds the gap between several of those rows. Their paired tests are large enough that reversal is improbable (the five-attribute collapse and the two-attribute floor are significant past 10^{-19}), but the F1 levels attached to them are not reproducible quantities.

The held-out separation has been broken a second time, less defensibly. Week 5 recorded that the evidence mask was derived from Amazon–Walmart failures, mitigated by the mask having an argument independent of any measurement and being inert on the selection dataset. The three-way split of Section 4 has neither mitigation: it was read directly off this dataset’s gold labels. It is reported as mechanism and excluded from the default configuration for that reason. Restoring the guarantee needs a third dataset used for nothing else.

Missingness is still absent from the feature vector. Section 4 shows the two kinds of blank carry opposite signal, and the six per-attribute features represent them identically. Encoding it was deliberately not attempted: it would be a further design choice fitted to the gold labels of the one dataset where the method shows no effect. It is deferred to a held-out protocol or a fourth dataset.

7 Next

1. **A held-out protocol.** Every design choice in this work — the selection rule, the mask, its three-way split — was read on the data it is reported over. Fixing choices on one half and reporting on the other would strengthen the whole thesis more than any further tuning.
2. **Provider-conditioned analysis.** With the provider now recorded per call, the next repeat experiment can report the answer distribution per provider rather than pooled, which would separate how much of the 5.2% flip rate is routing and how much is sampling inside one provider. It also makes pinning a single provider a testable way to buy reproducibility, at whatever cost in availability that turns out to carry.
3. **Sampler sweep.** All Step 1.5 runs use the **diversity** profile sampler, held fixed so the other variables could be read. Sweeping the five available samplers clears a stated limitation and widens rather than narrows the claim.